

Which one of the following properties is not of transition elements

- (a) Colour
- (b) Paramagnetism
- (c) Fixed valency**
- (d) None of the above

Answer: (c) Fixed valency

Explanation: The correct choice is Fixed valency. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

162. In which of the following, tendency towards formation of coloured ions is maximum

- (a) s-block elements
- (b) d-block elements**
- (c) p-block elements
- (d) f-block elements

Answer: (b) d-block elements

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

163. The electronic configuration $1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^6$ corresponds to

- (a) Mn^{2+}
- (b) Fe^{2+}**
- (c) Co

(d) Ge

Answer: (b) Fe^{2+}

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

164. Which of the following statements is not correct about the electronic configuration of gaseous chromium atom

- (a) It has 5 electrons in 3d and one electron in 4s orbitals
- (b) The principal quantum numbers of its valence electrons are 3 and 4
- (c) It has 6 electrons in 3d orbital**
- (d) Its valence electrons have quantum number 'l' 0 and 2

Answer: (c) It has 6 electrons in 3d orbital

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

165. Zn and Hg belong to the same group, they differ in many of their properties. The property that is shared by both is

- (a) They form oxide readily**
- (b) They react with steam readily
- (c) They react with hot concentrated sulphuric acid
- (d) They react with hot sodium hydroxide

Answer: (a) They form oxide readily



Explanation: The correct choice is They form oxide readily. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

166. Which of the following ionic species will impart colour to an aqueous solution

- (a) Ti^{4+}
- (b) Cu^{+}
- (c) Zn^{2+}
- (d) Cr^{3+}**

Answer: (d) Cr^{3+}

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

167. The number of electrons in the outermost shell of the 3d-transition elements generally remains

- (a) $(n-1)d^n$
- (b) nd^n
- (c) ns^2**
- (d) $(n-1)s^2$

Answer: (c) ns^2

Explanation: The correct choice is ns^2 . This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

168. The 3d-elements show variable oxidation states. What is the maximum oxidation state shown by the element Mn

- (a) + 4
- (b) + 5
- (c) + 6
- (d) + 7**

Answer: (d) + 7

Explanation: Transition elements can use both ns and $(n-1)d$ electrons in bonding because these levels are close in energy, so several oxidation states are possible.

169. Which of the following ions gives coloured solution

- (a) Cu^{+}
- (b) Zn^{++}
- (c) Ag^{+}
- (d) Fe^{++}**

Answer: (d) Fe^{++}

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

170. Which metal represents more than one oxidation state

- (a) Al
- (b) Na
- (c) Mg
- (d) Fe**



Answer: (d) Fe

Explanation: Assign oxidation numbers from charge balance and the accessible $ns/(n-1)d$ valence electrons of the transition element.

171. A reduction in atomic size with increase in atomic number is a characteristic of elements of

- (a) High atomic masses
- (b) d-block
- (c) f-block**
- (d) Radioactive series

Answer: (c) f-block

Explanation: The correct choice is f-block. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

172. Which one of the following characteristics of the transition metals is associated with their catalytic activity

- (a) Variable oxidation states**
- (b) High enthalpy of atomization
- (c) Paramagnetic behaviour
- (d) Colour of hydrated ions

Answer: (a) Variable oxidation states

Explanation: Transition metals are effective catalysts because they can change oxidation state and form intermediate complexes/adsorb reactants on their surfaces.

173. The number of oxidation states of manganese is

- (a) 4
- (b) 6**
- (c) 3
- (d) 8

Answer: (b) 6

Explanation: Assign oxidation numbers from charge balance and the accessible $ns/(n-1)d$ valence electrons of the transition element.

174. Which of the following transitional metal has lowest density

- (a) Sc**
- (b) Ti
- (c) V
- (d) Cr

Answer: (a) Sc

Explanation: Compare the relevant atomic masses and/or densities; the listed correct option has the greatest value among the choices.

175. Which of the following transitional metal has lowest boiling point (B.P.)

- (a) Zn**
- (b) Sc
- (c) Ti
- (d) V

Answer: (a) Zn



Explanation: Melting/boiling trends depend largely on the strength of metallic bonding, which is influenced by the number of valence d and s electrons available for bonding.

176. Which of the following electronic configuration is that of a transitional element

- (a) $1s^2, 2s^2p^6, 3s^2p^6d^{10}, 4s^2p^6$
- (b) $1s^2, 2s^2p^6, 3s^2p^6d^{10}, 4s^2p^1$
- (c) $1s^2, 2s^2p^6, 3s^2p^6d^2, 4s^2$**
- (d) $1s^2, 2s^2p^6, 3s^2p^6, 4s^2$

Answer: (c) $1s^2, 2s^2p^6, 3s^2p^6d^2, 4s^2$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

177. Fe has been placed in the eighth group, the number of electrons in the outermost orbit is

- (a) 1
- (b) 2**
- (c) 3
- (d) 4

Answer: (b) 2

Explanation: The correct choice is 2. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

178. Due to covalent bonding, the transitional metals are

- (a) Lustrous
- (b) Conductor
- (c) Hard and brittle**
- (d) Ductile

Answer: (c) Hard and brittle

Explanation: The correct choice is Hard and brittle. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

179. The magnetic moment of a metal ion of first transition series is 2.83 BM. Therefore it will have unpaired electrons

- (a) 6
- (b) 4
- (c) 3
- (d) 2**

Answer: (d) 2

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

180. Which of the following pair of ions may exhibit same colour

- (a) Cr^{+++} and Fe^{++}**
- (b) Ti^{+++} and V^{++}
- (c) Fe^{+++} and Mn^{++}
- (d) Cu^{+} and Ni^{++}



Answer: (a) Cr^{+++} and Fe^{++}

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

